Gustavo Landfried
Chemin des Bégonias 1
1018 Lausanne

Avril 2026

Madame, Monsieur,

Je suis citoyen suisse domicilié à Lausanne, docteur en Informatique spécialisé en inférence bayésienne causale et licencié en Sciences Anthropologiques. Je me suis installé en Suisse romande en avril 2026 après plusieurs années en Argentine, où j'ai développé mes activités de recherche, d'enseignement et de conseil. Je travaille actuellement comme Senior Data Scientist à distance pour des multinationales latino-américaines. Cette combinaison de formations n'est pas accidentelle: c'est une trajectoire construite délibérément pour opérer à l'intersection entre la rigueur statistique et la compréhension des phénomènes sociaux, économiques et technologiques.

Au cours des dernières années, j'ai dirigé des projets statistiques avec de nombreux interlocuteurs institutionnels. Pour la *Foundation for Innovative New Diagnostics* (FIND) et Fiocruz, j'ai conduit l'évaluation statistique de tests diagnostiques pour la maladie de Chagas dans quatre pays à l'aide de modèles B-LCA; ce faisant, j'ai détecté une erreur méthodologique dans le modèle du consortium européen COST Action CA18208, ce qui a conduit à une publication dans *PLoS Neglected Tropical Diseases* (2024). J'ai développé des modèles bayésiens hiérarchiques pour des décisions de politique commerciale chez Coca-Cola FEMSA et KAVAK. Mes publications dans le *Journal of Statistical Software* (2025), le *Journal of Computational Social Science* (2025) et *PLoS ONE* (2019) reflètent la capacité de combiner rigueur méthodologique et résultats communicables. Je maîtrise R (Stan), Python (PyMC, Pyro, PyTorch), Julia (Turing) et les architectures de données à grande échelle (Delta Lakehouse, Spark).

En tant qu'enseignant universitaire d'Inférence Bayésienne Causale (UNSAM/UBA, 2024–actuel), j'ai élaboré un programme original de huit unités allant de la spécification de modèles graphiques causaux jusqu'à la prise de décision ergodique. En tant que fondateur et coordinateur de Bayes Plurinacional, j'organise des événements scientifiques annuels dans plusieurs pays d'Amérique latine (Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026). J'ai dirigé 7 mémoires de licence en Informatique et Science des données à l'Université de Buenos Aires.

L'inférence causale bayésienne est l'approche qu'il faut pour quantifier l'incertitude et distinguer les corrélations des effets causaux réels. Elle a des applications directes dans de nombreux domaines: santé publique, statistique et prospective institutionnelle, décisions d'affaires avec peu de données, recherche académique et formation spécialisée. Je cherche en Suisse romande parce que c'est la région où je m'installe avec mon épouse (nous nous sommes mariés à Lausanne il y a 15 ans) et où nous souhaitons construire notre vie de façon durable. Je suis disponible immédiatement, sans nécessité de déménagement.

Sincèrement,

LANDFRIED

Gustavo Landfried

Analyse causale des données

Landfried, Gustavo

Nationalité	Suisse - Argentine
Langues	Français C1 | Anglais C1 | Espagnol maternelle | Allemand A1
Résidence	Lausanne (Suisse) | Buenos Aires (Argentine)
Date de naissance	Octobre 1985
Téléphone	+41 77 407 1874 | +54 911 5103 7334
Email	gustavolandfried@gmail.com
Web	glandfried.github.io/
Linkedin	linkedin.com/in/gustavo-landfried
Github	github.com/glandfried

Doctorat en informatique | Anthropologue social | **Inférence bayésienne causale**

- Responsable de projets statistiques au sein d'organisations inter-et multinationales
- Chercheur avec des publications en inférence bayésienne causale (JSS, JCSS, PLoS ONE)
- Fondateur et coordinateur de la communauté statistique bayésienne d'Amérique latine
- Développeur de logiciels statistiques sur-mesure adaptés à chaque problème (Pyro, PyMC)
- Spécialiste des méthodes statistiques efficaces pour la détection de variables cachées
- Flexibilité interdisciplinaire et communication accessible pour des publics variés

Professionnel des statistiques au sein d'organisations internationales.

2025 – actuel

Causal Data Scientist @ *Amérique latine*

○ **Coca-Cola (FEMSA)**. Analyse des effets causaux hétérogènes dans le système de promotions de fidélisation Juntos+. Problème: les promotions étaient conçues selon des règles fixes. Résultat: développement de la stratégie adoptée par l'équipe, fondée sur un réseau bayésien causal.
○ **Marketplace automobile (KAVAK)**. Évaluation de l'impact des efforts de marketing. Problème: A/B Testing avec une faible signification statistique et attribution par règle fixe. Résultat: une méthodologie efficace en termes de données pour l'analyse d'expériences causales et l'attribution Data-Driven.
○ **Optimiseur de médias payants (Muttdata)**. Répartition du budget entre les canaux publicitaires. Problème: mauvaises performances prédictives pour les conversions de qualité avec faible fréquence. Résultat: un modèle bayésien hiérarchique qui a considérablement amélioré les prédictions d'impact.

2023 – 2024

Responsable de l'évaluation des tests de diagnostic @ *Amérique latine et Europe*

○ **Foundation for Innovative New Diagnostics (FIND)**. Évaluation du test de diagnostic rapide de la maladie de Chagas en Amérique du Sud. Problème: la méthodologie actuelle, coûteuse et lente, constitue un goulot d'étranglement. Résultat: nous avons montré que les tests rapides permettent de développer une méthodologie plus efficace et plus sûre.
○ **European Cooperation in Science and Technology (Action CA18208)**. "Novel tools for test evaluation and disease prevalence estimation". Problème: j'ai signalé une erreur qu'ils commettaient dans l'utilisation du modèle de Dendukuri. Résultat: publication dans *PLoS Neglected Tropical Diseases*.
○ **Administration Nationale de la Santé (Malbrán)**. Expérience sur l'impact des souches de Chagas sur les diagnostics. Problème: faible signification statistique en raison du peu de données. Résultat: signification élevée grâce à un modèle bayésien causal; publication *Journal of Molecular Diagnostics*

2022 – actual

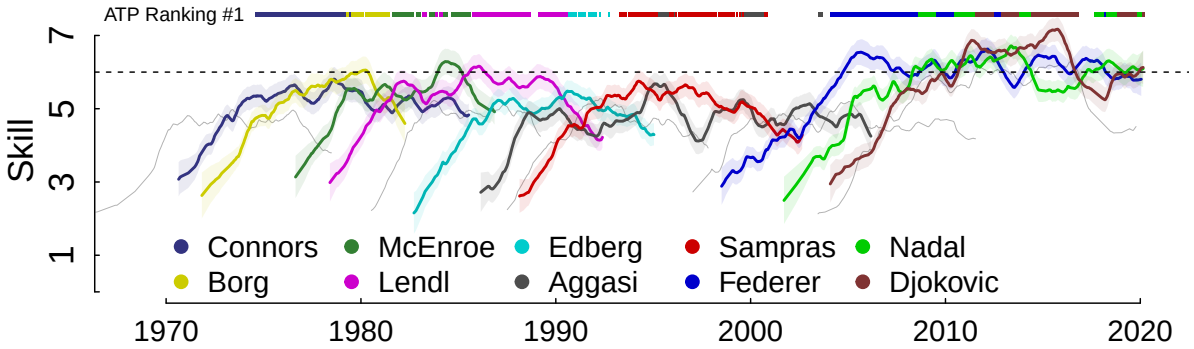
Promoteur de la communauté continentale de statistique bayésienne @ *Amérique latine*

○ **Bayes Plurinational**. Problème: l'adoption de l'inférence bayésienne est en retard au niveau continental. Résultat: organisation ininterrompue d'événements d'intelligence artificielle et formation de ressources humaines en statistique bayésienne, Santiago 2023, Salta 2024, Bogota 2025, Ciudad del Este 2026.
○ **Formation professionnelle en inférence bayésienne causale**. Distinguer la corrélation de la causalité et trouver les décisions optimales au fil du temps. Voir le contenu complet dans Enseignement.

Système de raisonnement optimal en contexte d'incertitude

2021 – 2022 Estimateur de compétences de pointe @ *Journal of Statistical Software*

Inférence efficace dans des réseaux causaux avec des millions de nœuds et des structures irrégulières à travers des algorithmes de passage de messages et des méthodes d'approximation analytique.



2019 – 2020 Impact du réseau social sur la compétence @ *Journal of Computational Social Sicence*

Étude de l'impact de la structure sociale sur la compétence des individus. Problème: les études socioculturelles classiques ne pouvaient pas évaluer cet impact. Résultat: les données massives de la communauté OGS (Online Go Platform), l'analyse de graphes (popularité, proximité, intermédiarité) et l'estimateur de compétence de pointe ont permis de l'évaluer.

2017 – 2018 Impact de la formation des équipes sur la compétence @ *PLoS ONE*

Nous avons analysé une communauté du jeu Risk, entre 2006 et 2013, dans laquelle il était possible de jouer en équipe. Nous avons montré que jouer en équipe conduit à une compétence plus élevée à long terme, et que jouer avec un partenaire stable accélère l'acquisition de compétences.

Professionnel en anthropologie sociale avec une orientation quantitative

2015 – 2016 Programme d'analyse sociale de la citoyenneté latino-américaine @ *PASCAL*

Entité principale créée par la loi sur les médias en Argentine pour la mesure de l'audience (rating) de la télévision. Mon rôle au sein de PASCAL était double: coordinator entre l'équipe technique et l'équipe sociale, et administrateur de la base de données et du système d'enquêtes téléphoniques automatisées.

2014 Formation pour l'admission au doctorat en informatique @ *Universidad de Buenos Aires*

2012 – 2013 Union des travailleurs de l'économie populaire @ *Programme "Argentina Trabaja"*

Assistant l'ancien président de l'Institut national de technologie industrielle 2002-2011. Mon travail s'est concentré sur les coopératives de travail d'"Argentina Trabaja". Mes tâches étaient dirigées par Enrique Palmeiro, assistant de François, qui a été nommé Pape alors que je travaillais avec lui. Nous menions des évaluations des coopératives pour promouvoir leur production.

La société et la culture vues comme des systèmes complexes

2011 – 2012 Prédiction des alliances parlementaires @ *Université de Lausanne*

Cours à l'Institut de mathématiques appliquées de la Faculté des sciences sociales et politiques de l'Université de Lausanne sur la complexité et les modèles basés sur les agents, dispensé par Jorge Peña. Mémoire de licence mettant en œuvre le modèle de prédiction d'alliances de Robert Axelrod appliqué aux démocraties parlementaires européennes.

2010 Formation en sciences exactes et naturelles @ *Universidad de Buenos Aires*

2008 – 2009 Anthropologie, complexité, simulation et ethnographie @ *Grupo Antropocaos*

Assistant de recherche pour le projet UBACyT "Modèles de simulation et anthropologie, leur relation avec le travail ethnographique". Assistant d'enseignement pour le séminaire "Sociétés artificielles et ethnographie"

Education Université de Buenos Aires

2016 – 2022 Doctorat en **Informatique**

2012 – 2014 Diplôme universitaire (bac+5) en Informatique. (Suspendu après la promotion au doctorat)

2005 – 2009 Diplôme universitaire (bac+5) en **Sciences Anthropologiques**.

Enseignement

2024 – actuel *Inférence bayésienne causale* @ *Laboratorios de Métodos Bayesianos - ECyT UNSAM*

(1) Spécification et évaluation de modèles causaux alternatifs. (2) Inférence exacte et approchée (passage de messages, variationnel, MCMC, SBI). (3) Flux d'inférences, critère backdoor et Do-Calcul. (4) Algorithmes et frameworks Causal ML. (5) Algorithmes temporels et modèles temporels génératifs. (6) Inférence contrefactuelle et effets causaux directs. (7) Prise de décision, optimisation comme inférence et utilité ergodique. (8) Interventions socioéconomiques et instruments financiers productifs.

2019 – actuel *Mentorat individuel*. Directeur de 7 thèse de master en informatique et science de données.

2016 – 2024 Cours précédents à l'UBA (Informatique et Anthropologie): Inférence bayésienne, Algorithmes et structures de données (I et II), Introduction à l'informatique, Programmation fonctionnelle (`Haskell`), Sciences sociales computationnelles.

Recherche scientifique

Articles

○ **Landfried** G., Mocskos E. *TrueSkill Through Time: reliable initial skill estimates and historical comparability in Julia, Python and R*. **Journal of Statistical Software. 2025**. 10.18637/jss.v112.i06

○ **Landfried**, G., Cairo G., Mocskos E. *Network Position and Learning Dynamics: Unveiling the Impact of Social Structure on Skill Acquisition in Online Gaming Platforms*. **Journal of Computational Social Sicence. 2025** 10.1007/s42001-025-00370-2.

○ Longhi, SA; Muñoz-Calderón, A; García-Casares, L; Irazu, L; Rodríguez, MA; **Landfried**, G; Alonso-Padilla, J; Schijman, AG; and Chagas-group *Inter-Laboratory Harmonization Study and Prospective Evaluation of the PURE-T. cruzi-LAMP Assay for Detecting Parasite Presence in Newborn Dried Blood Spots*. **The Journal of Molecular Diagnostics. 2024.** 10.1016/j.jmoldx.2024.08.007

○ Denwood, M; Nielsen, S; Olsen, A; Jones, H; Coffeng, L; **Landfried**, G; Nielsen, M; Levecke, B; Thamsborg, SM; Eusebi, P; Meletis, E; Kostoulas, P; Hartnack, S; Erkosar, B; Toft, N. *All that glitters is not gold: an interpretive framework for diagnostic test evaluation using Ascaris lumbricoides as a conceptual example*. **Plos Neglected Tropical Disease. 2024**.10.1371/journal.pntd.0012481

○ **Landfried**, G; Fernandez Slezak, D; Mocskos, E: *Faithfulness-boost effect: Loyal teammate selection correlates with skill acquisition improvement in online games.* **PLoS one. 2019**. 10.1371/journal.pone.0211014

Qualités

S'il y a de l'équilibre, moins ça pèse.

Outils

`Delta Lakehouse` (`DataBricks`, `Spark`, `Streaming`, `Pipeline` ...), `Agentic AI` (`Claude Code`, `MCP`, `Hooks`, `Subagents`, `Skills`, ...), `Python` (`Pytorch`, `TensorFlow`, `Keras`, `PyMC`, `Pyro`, `Particles`, `Scipy`, `Sklearn`, `Pandas`, `Numpy`, ...), `R` (`Stan`, `Tidyr`, ...), `Julia` (`Turing`, ...), `C++` (`MPI`), `C#` (`Infer.NET`, ...), `Java`, `Haskell`, `Bash` (`screen`, `ssh`, `vi`, `rsync`, `awk`, ...), `SQL`, `NoSQL`, `Git`, `Docker`, `Latex` (`Tikz`), `Html`, ...

REPÚBLICA ARGENTINA

Universidad de Buenos Aires

Facultad de Ciencias Exactas y Naturales

El Rector de la Universidad y el Decano de la Facultad

Por cuanto, el Licenciado en Ciencias Antropológicas (orientación Sociocultural) **Gustavo Andres Landfried** natural de la Ciudad de Buenos Aires, ha finalizado el 9 de abril de 2024 los estudios correspondientes al **Doctorado de la Universidad de Buenos Aires, área Ciencias de la Computación.**

Por lo tanto, de acuerdo con las normas vigentes en esta Universidad, le confieren el presente diploma de **Doctor de la Universidad de Buenos Aires, área Ciencias de la Computación.**

Buenos Aires, 20 de noviembre de 2024

Documento Nacional de Identidad 31916400

Título de la tesis: "Análisis bayesiano del aprendizaje en comunidades de video juegos."

Calificación Sobresaliente aprobada el 09/04/2024

LEONARDO FEDERICO GOTTIFREDI
DIRECTOR GENERAL DE TÍTULOS Y PLANES

MARÍA CATALINA NOSIGLIA
SECRETARIA DE ASUNTOS ACADÉMICOS

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Departamento de Certificación de Títulos

Landfried, Gustavo Andres
DNI - 31916400
UBA - 294/21

Diploma

Ministerio de Capital Humano
Secretaría de Educación
Subsecretaría de Políticas Universitarias
Argentina

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REPÚBLICA ARGENTINA

Universidad de Buenos Aires

Facultad de Filosofía y Letras

El Rector de la Universidad y la Decana de la Facultad

Por cuanto, **Gustavo Andrés Landfried**

natural de la Ciudad de Buenos Aires, ha finalizado el 20 de mayo de 2013 los estudios correspondientes a la carrera de **Ciencias Antropológicas.**

Por lo tanto, de acuerdo con las normas vigentes en esta Universidad, le confieren el presente diploma de **Licenciado en Ciencias Antropológicas (orientación Sociocultural).**

Buenos Aires, 12 de agosto de 2014

SOFÍA IRENE THISTED
Secretaria de Asuntos Académicos

GRACIELA ALEJANDRA MORGADE
DECANA

ALBERTO E. BARBIERI
RECTOR

MARÍA CATALINA NOSIGLIA
SECRETARIA DE ASUNTOS ACADÉMICOS

Documento Nacional de Identidad 31916400

LEONARDO FEDERICO GOTTIFREDI
DIRECTOR GENERAL DE TÍTULOS Y PLANES

MINISTERIO DE EDUCACIÓN

DIRECCIÓN NACIONAL DE GESTIÓN UNIVERSITARIA

Certifico que las firmas del Rector, Dr. Alberto Edgardo BARBIERI, Secretaria de Asuntos Académicos, Lic. María Catalina NOSIGLIA y el Director General de Títulos y Planes, Sr. Leonardo Federico GOTTIFREDI que anteceden, guardan similitud con la/s que obran en nuestros registros.

Buenos Aires, - 3 NOV 2014

Cont. MARIA I. ILLAPRESCAPE
DIRECCIÓN NACIONAL DE GESTIÓN UNIVERSITARIA

LANDSRIEDA, GUSTAVO (21)

REPUBLICA ARGENTINA
Ministerio de Educación
Presidencia de la Nación
ME000000620328

Firma del interesado

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DNI 31916400
UBA - Licenciado en Ciencias Arts

Bayesian Causal Inference

IBC1.2026.1

Professor: Gustavo Landfried[1,2,3]

1. School of Science and Technology, Universidad Nacional de San Martín (UNSAM)
2. Bayesian Methods Laboratory
3. Muttdata

Contact: `glandfried@dc.uba.ar` `glandfried.github.io`

April 13, 2026

All data-driven sciences develop causal arguments to explain and predict the world. The evaluation of causal hypotheses is attracting increasing interest from industry, which needs to measure the real impact of its actions. The advantage of causal models lies in their predictive capacity, which adapts naturally to contextual changes such as human interventions. Moreover, we know that if causal arguments correspond to the underlying causal reality, no artificial intelligence model, however complex, can outperform them.

In this course we will review the foundations for evaluating alternative causal models M given data D, $P(M|D)$, both with and without interventions. We will also review methods for generating predictions of hypotheses H internal to models M, $P(H|D, M)$, both for evaluating out-of-sample causal effects and for predicting the impact of alternative counterfactual actions. Finally, we will address the problem of optimal decision-making as an inference problem in which we verify that none of the alternative counterfactual decisions improve the result on the target variable.

Faced with the dizzying pace of AI tools, this course prioritizes immutable foundations. Despite all advances, since the 18th century no new system for reasoning under uncertainty has been proposed. If machines ever surpass humans, they will have to do science by strictly applying probability rules to evaluate causal theories. To reach (intersubjective) truths under uncertainty, one simply needs to know how to apply and preserve the universal principle of not lying: do not assert more than is known, without hiding what is known. This yields optimal belief distributions given the available information, unimprovable in practical terms.

Objectives

This course focuses on evaluating alternative causal arguments through the (approximation to) strict application of probability rules (Bayesian approach), the system of reasoning under uncertainty. The main objective is to review methods developed in recent decades for:

- **Causal models.** Mathematically specifying causal arguments expressed in natural language through intuitive graphical methods.
- **Inference flow.** Understanding when inference flow opens and closes in causal structures, in order to separate spurious influence from causal influence.
- **Causal effect estimation.** Applying Machine Learning methods to identify the causal effect between pairs of variables from observational data without interventions.
- **Counterfactual predictions.** Specifying joint probability distributions between factual and counterfactual outcomes, and generative reasoning about counterfactual consequences.
- **Decision-making over time.** Selecting optimal decisions in action-perception cycles with a hidden (simulated) nature.

About the instructor

Gustavo Landfried is a professor of Bayesian Causal Inference at the School of Science and Technology of the Universidad Nacional de San Martín (UNSAM), and one of the leading figures in Bayesian causal inference education in Latin America. After obtaining a degree in Social Anthropology, Gustavo earned his PhD in Computer Science from the Universidad de Buenos Aires (UBA) under the supervision of the directors of the Applied Artificial Intelligence Laboratory (Diego Slezak) and the Interdisciplinary High-Performance Computing Laboratory (Esteban Mocskos). He specialized in Bayesian skill analysis for the video game industry, which led him to develop and maintain the state-of-the-art packages in Python, Julia and R (*TrueSkill Through Time*, Journal of Statistical Software, 2025). Gustavo currently works in industry (Muttdata) as a Data Scientist developing Bayesian and causal inference projects for large multinational corporations in Latin America. He is also co-founder and coordinator of the community `bayesplurinacional.org`, organizing annual scientific events across the continent (Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026). He has supervised seven BSc/MSc theses in Computer Science and Data Science.

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All sub-units include assessed assignments. Upon completion of each full unit, an evaluation is conducted based on multiple-choice questions with belief distributions.

1 Specification and Evaluation of Causal Arguments

System of reasoning under uncertainty, graphical specification of causal arguments, and evaluation of alternative causal models with and without interventions.

1.1 Alternative Causal Arguments and Uncertainty

Truth and falsehood. Intersubjective agreements: maximum uncertainty given available information. Honest belief distribution across parallel universes of causal arguments. Preservation of prior belief compatible with the data and prediction as a sum of parallel universes compatible with what one wants to predict. System of reasoning under uncertainty: the rules of probability. Graphical methods for mathematically specifying probabilistic causal arguments: Bayesian networks. Basic properties of graphical models: structure, independence and factorization. Measuring uncertainty over hypothesis spaces internal to models, $P(\text{Hypothesis}|\text{Data}, \text{Model})$. Evaluating alternative causal arguments from observational data without interventions, $P(\text{Model}|\text{Data})$. The evidence, $P(\text{Data}|\text{Model})$, and its transformations (geometric mean and logarithm).

Assignment 1.1: Specification and evaluation of Monty Hall and Base models.

1.2 Surprise: The Problem of Communication with Reality

The fundamental elements of the reasoning system under uncertainty (Kolmogorov and Rényi axioms). The invariant structure of empirical data. Its isomorphism with communication systems with reality. Surprise as a source of information. The equivalence between the geometric

mean of the prediction, $P(\text{Data}|\text{Model})$, and cross-entropy. The advantage of causal models that correspond to the underlying causal reality. The mathematical definition of the principle of not lying: maximum entropy given available information. Specification of causal models for contexts with corrupted data. Context-adaptive causal mechanisms: do-operator and gates. Specification of causal arguments through factor graphs. Identification of causal models with and without interventions. Evaluation of alternative measurement procedures.

Assignment 1.2: Specification and evaluation of the Monty Hall model with corrupted data.

2 Inference Methods and Probabilistic Programming

Exact inference methods through message passing between nodes of the causal network, and approximate inference — both analytic and sampling-based — through probabilistic programming languages.

2.1 Exact Inference and Message Passing

Exact inference. The variable elimination algorithm. Decomposition of probability rules as messages exchanged between nodes of causal networks: sum-product algorithm. The extended Monty Hall model case. Conjugate distributions. Beta-Bernoulli model and its applications in medicine and marketing. Univariate Gaussian models and the exact solution to the skill estimation model. Multivariate Gaussian models: posterior and evidence. Bayesian linear regression. Arbitrary hypothesis selection via maximum likelihood. Interpretation of estimated parameters under known causal realities. Evaluation of linear models of increasing complexity. Absence of overfitting through strict application of probability rules. Prediction based on model ensembles, $P(\text{Data})$.

Assignment 2.1: Evaluation of linear models of increasing complexity.

2.2 Forgetting: The Problem of Approximate Inference

Analytic approximation methods. KL divergence and approximate inference via expectation propagation. The case of skill estimation in the video game industry. Reversed KL divergence and approximate inference via variational inference. The Bayesian version of the recommendation system that won the Netflix competition. The probabilistic programming language `Pyro`. Sampling-based approximation methods. Monte Carlo approximate inference: Metropolis-Hastings, Hamiltonian Monte Carlo and Sequential Monte Carlo. Probabilistic programming languages `PyMC`, `Stan`, `Turing`. The case of evaluating alternative measurement instruments and the Bayesian p-value. Generative approximation methods. Approximate Bayesian Computation (ABC): likelihood-free inference via summary statistics and distance metrics. Current methods for estimating the joint distribution via neural networks from simulated data (SBI) and efficient inference.

Assignment 2.2: Evaluation of alternative measurement instruments: Chagas diagnostic test with corrupted data.

3 Causal Predictions

Elimination of spurious influence from predictions and the steps of causal inference: 1) specification of the causal structure, 2) identification of the estimand, 3) estimation of the causal effect.

3.1 Inference Flow and Elimination of Spurious Influence

Inference flow in causal networks. Opening and closing of inference flow in basic structures: pipe, fork and collider. Opening and closing of inference flow along paths of any length: d-separation. Elimination of spurious influence from predictions: backdoor controls. Good and bad controls. Neutral controls and heterogeneous causal effects. Use of linear regression for estimating non-linear causal effects. Interpretation of estimated coefficients. The Table 2 Fallacy. The steps of causal inference: 1) specification of the causal structure, 2) identification of the estimand, 3) estimation of the causal effect. Discovery of causal structures compatible with estimated joint probability distributions from data.

Assignment 3.1: Causal predictions under diverse underlying causal realities.

3.2 Estimands and Do-Calculus

The estimand for the general causal effect: adjustment formula and propensity score. The fundamental characteristic of randomized experiments: independence between treatment and its potential counterfactual outcomes. The "conditional ignorability" analogy as a criterion for selecting control variables in the potential outcomes approach to causal inference. Estimand of causal effects under conditional ignorability and deterministic consistency. The 3 rules of Do-Calculus: Rule 1, Ignoring an observation; Rule 2, Intervention as observation; Rule 3, Ignoring an intervention. Alternative estimand when hidden controls exist: frontdoor formula. Alternative estimand when randomized experiments fail: instrumental variables.

Assignment 3.2: Estimation of causal effects: general, frontdoor and instrumental.

4 The Algorithm Zoo

From early strategies to state-of-the-art supervised learning models with proven capacity for estimating heterogeneous causal effects in high-dimensional settings.

4.1 Traditional Methods

Traditional effect estimation methods. Comparison of treated and control units with similar characteristics: Matching. Use of the Propensity Score as a similarity measure. Reweighting the sample through the Propensity Score: Inverse Probability Weighting (IPW). Limitations of traditional methods in high-dimensional settings. The Causal Machine Learning (CausalML) solution: high-capacity predictive algorithms for estimating heterogeneous causal effects (CATE). The Python causal inference ecosystem: integration of `DoWhy`, `EconML` and `CausalML`. The concept of Meta-Learners: strategies for decomposing estimation of the conditional causal effect (CATE) into standard supervised prediction problems. S-Learner and T-Learner: advantages and disadvantages. The impact of lack of "positivity" on causal predictions: the personal lending industry case.

Assignment 4.1: Causal predictions under different degrees of positivity.

4.2 Leading Algorithms in Causal Inference Competitions

Fundamentals and application of the Doubly Robust Learner. Causal inference competitions at the Atlantic Causal Inference Conference. The regularization problem in causal effect estimation. Estimation strategy under partially linear models: orthogonalization. Fundamentals and application of Doubly Machine Learning: regularization in nuisance models and isolation of the causal effect. Non-linear algorithms based on tree ensembles. From Random Forests to Causal Forests. From Bayesian Additive Random Trees to Bayesian Causal Forests. The problem of estimated confidence intervals in data competitions and possible solutions. Hybrid algorithms.

Assignment 4.2: Causal inference competition ACIC 2022.

5 Time

Specialized algorithms for estimating causal effects in time series. Generative temporal causal models: specification, inference and evaluation.

5.1 Temporal Algorithms

The advantage of longitudinal data: time-invariant fixed effects. Classical Difference-in-Differences (DiD). The problem of DiD with heterogeneous effects. DiD with multiple periods and staggered adoption. New robust estimators: Callaway-Sant'Anna, Sun-Abraham, Doubly Robust DiD. The problem of absence of comparable control units. Construction of synthetic control groups through weighted combination of untreated units. Combining synthetic weights with DiD structure allowing staggered timing and heterogeneous trends: Synthetic Difference-in-Differences. Generalization of synthetic controls for multiple simultaneously treated units via missing data imputation: Matrix Completion.

Assignment 5.1: Estimation of temporal causal effects comparing different methods.

5.2 Generative Temporal Causal Models

Specification of temporal causal models and their unrolling. Skill estimation in the video game and sports industries. Inference flow in complete-history generative models: forward and backward propagation. Undefined messages and convergence via Loopy Belief Propagation. The forward-backward algorithm (smoothing): skill estimation over time. Temporal predictions and leave-one-out predictions: costs. Skill estimation in the history of tennis. Interpretation of odds offered by bookmakers. Simplified betting game: the honest coin. Properties of betting games. The advantage of individual diversification (negative entropy).

Assignment 5.2: Specification of a generative temporal causal betting model. Simulation and impact analysis of cooperation and individual saving.

6 Counterfactuals

Specification of models with counterfactuals, equivalence between causal inference approaches, and decomposition of total effects into natural direct and indirect effects via the mediation formula.

6.1 Models with Counterfactuals and Equivalence between Causal Inference Approaches

The levels of causal reasoning. Specification of causal models with counterfactuals: Twin Networks. Prediction of counterfactuals using factual information: Monty Hall. Revisiting the equivalence of the two causal inference approaches: Potential Outcomes and Structural Causal Models. Compliance analysis of ignorability via Twin Networks: examples. Specific constraints of Structural Causal Models: one exogenous variable per endogenous variable and determinism of endogenous variables. Breaking of Ignorability-Backdoor equivalence in general causal generative models: the paradigmatic counterexample. Generalization of the Backdoor causal inference approach to general causal generative models.

Assignment 6.1: Predicting counterfactual values using factual information.

6.2 Using Counterfactuals to Estimate Direct Causal Effects

Total vs. Direct and Indirect causal effects. Conditional Direct Effect: a contraindicated intervention. Decomposition of the total effect: Natural Direct Effect, the impact produced by treatment under an untreated mediator; Natural Indirect Effect, the impact produced by the change in the mediator under treatment. The Burks-Hammel debate: measuring gender discrimination in university admissions. Identifiability of natural effects: elimination of spurious influence between the mediator and the outcome. Mediation formula: NDE and NIE estimands from observational data and structural models. Estimation of NDE and NIE from observational data without interventions.

Assignment 6.2: Evaluation of discrimination in simulated data.

7 Decision-Making

The probability-evolution isomorphism, consequences of the multiplicative nature of variant selection, and the temporal correction of expected utility theory.

7.1 Decision-Making Over Time: The Epistemic-Evolutionary Cost Function

Von Neumann's expected utility theory. John L. Kelly's critique of Von Neumann. Betting games. The structural equivalence between betting games, hypothesis evaluation in probability, and natural selection in evolution. The advantage of individual diversification (negative entropy). The emergent properties of material forms and the advantage of variance reduction through cooperation. The fallacy of the prisoner's dilemma and the tragedy of the commons. The advantage of cooperative specialization. The Kelly criterion: cooperating with oneself. Fractional Kelly.

Assignment 7.1: Skill estimation and tennis betting.

7.2 Temporal Correction of Expected Utility Theory and Optimization as Inference

Systems of interaction with reality: action-perception. Active learning. Prediction by maximum uncertainty and action selection by minimum uncertainty. The ergodic (temporal) correction of expected utility theory. Optimal factual decision as absence of superior counterfactual alternatives. Optimization as inference. Counterfactual-based influence diagrams: the betting game and solution to specialization (individual and 2-person pool). Deterministic policies and random drip experiments. The personal lending industry case: deterministic credit allocation. Strategies for conducting micro-randomized experiments, minimizing data production and collection: the value of information.

Assignment 7.2: Micro-experiments and policy evaluation in the personal lending industry.

8 Socioeconomic Interventions

History of the colonial-modern turn, governance of the commons, analysis of current cases, and design of productive financial instruments at all levels (savings, land use planning, basic services, housing access, cycle reproduction).

8.1 General History, Commons and Successful Institutions

Limits of estimating causal effects between pairs of variables. The case of gender violence and random allocation of housing subsidies. The development of culture and agriculture. The 6 innovation centers of humanity. The colonial-modern turn: feudal pandemic, American silver, and Chinese technology. The era of genocides and the loss of cultural diversity. The deepening of the current ecological crisis and the inability of modern metropolitan science to resolve it. Characteristics of successful institutional systems in the management of the commons. Egypt: land distribution, agricultural productivity, population density and demographic dynamics. Argentina: land distribution, agricultural productivity, population density and demographic dynamics.

Assignment 8.1: Report: analysis of Argentina and Egypt data; estimates, conclusions and action proposals.

8.2 Productive Financial Instrument

Financial-productive systems in present-day Argentina: interest rate, risk and liquidity. Source of real value: water, land and life. Productive savings financial instrument — sowing pool — based on intensive agriculture and rurban development. The three actors of the system: savers, residents and administration. Savings based on pools for purchasing rural land (blocks of more than 100 ha) and interest rates aligned with rural rental values. General land planning: roads, housing, market-garden agriculture and common spaces. Specific planning of sectors and minimum size of rural areas (plots of 2000 m^2) and dwellings (250 m^2). Development of basic services: water, sewage, foundations and common area. Requirements for residents to acquire ownership: total cost and timelines for debt cancellation. The

complete cycle for process reproduction. Estimated capabilities and speed of potential growth.

Assignment 8.2: Writing the technical report and executive summary of the productive financial instrument.

Core Bibliography

- Koller, D. & Friedman, N. (2009). *Probabilistic Graphical Models.* MIT Press.
- MacKay, D. (2003). *Information Theory, Inference and Learning Algorithms.* Cambridge University Press.
- Jaynes, E.T. (2003). *Probability Theory: The Logic of Science.* Cambridge University Press.
- Pearl, J. (2009). *Causality: Models, Reasoning and Inference.* Cambridge University Press.
- Pearl, J. & Mackenzie, D. (2018). *The Book of Why.* Basic Books.
- Peters, J., Janzing, D. & Schölkopf, B. (2017). *Elements of Causal Inference.* MIT Press.
- Bishop, C. (2006). *Pattern Recognition and Machine Learning.* Springer.
- McElreath, R. (2020). *Statistical Rethinking.* CRC Press.
- Molak, A. (2023). *Causal Inference and Discovery in Python.* Packt.

Probabilistic Programming Languages

- `Pyro` (Uber AI): https://pyro.ai
- `PyMC`: https://www.pymc.io
- `Stan`: https://mc-stan.org
- `Turing.jl` (Julia): https://turinglang.org